Risks and Benefits of Anticoagulant and Antiplatelet Medication Use before Cataract Surgery

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Objective: To estimate the risks and benefits associated with continuation of anticoagulants or antiplatelet medication use before cataract surgery.

Design: Prospective cohort study.

Participants: Patients 50 and older scheduled for 19,283 cataract surgeries at nine centers in the United States and Canada between June 1995 and June 1997.

Intervention: None.

Main Outcome Measures: Intraoperative and postoperative (within 7 days) retrobulbar hemorrhage, vitreous or choroidal hemorrhage, hyphema, transient ischemic attack (TIA), stroke, deep vein thrombosis, myocardial ischemia, and myocardial infarction.

Results: Before cataract surgery 24.2% and 4.0% of patients routinely used aspirin and warfarin, respectively. Among routine users, 22.5% of aspirin users and 28.3% of warfarin users discontinued these medications before surgery. The rates of stroke, TIA, or deep vein thrombosis were 1.5/1000 among those who did not use aspirin or warfarin and 3.8/1000 surgeries among routine users of aspirin and warfarin who continued their medication before surgery. The rate was 1 event per 1000 surgeries among those who discontinued aspirin use (relative risk = 0.7, 95% confidence interval = 0.1–5.9). There were no events among warfarin users who discontinued use. The rates of myocardial infarction or ischemia were 5.1/1000 surgeries (aspirin) and 7.6/1000 surgeries (warfarin) among routine continuous users and no different from those of routine users who discontinued use.

Conclusions: The risks of medical and ophthalmic events surrounding cataract surgery were so low that absolute differences in risk associated with changes in routine anticoagulant or antiplatelet use were minimal. *Ophthalmology 2003;110:1784–1788 © 2003 by the American Academy of Ophthalmology.*

An issue of controversy is whether to stop anticoagulants or antiplatelet agents before cataract surgery.[1] Patients taking such agents may be at higher risk for ocular hemorrhages, but if such therapy is halted before surgery, patients may increase their risk of medical events for which the therapy was indicated. Few studies have examined the issue of whether to stop anticoagulant or antiplatelet therapy before ocular surgery, and many of these studies had a small sample size or lacked a control group.[2–10] The use of anticoagulant or antiplatelet therapy is common, especially among elderly patients, as it has been shown to reduce the risk of thromboembolic complications in patients with many coronary and vascular conditions.[11–17] Hence, there is a question of whether the risk of adverse medical events associated with temporarily stopping anticoagulant or antiplatelet therapy before cataract surgery outweighs the benefits of fewer hemorrhagic events. Because the overall risks of complications from cataract surgery are very low, it is difficult to determine whether brief cessation of anticoagu-

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*Clinical sites, investigators, and Data Safety and Monitoring Board members are cited in Schein et al.[19]

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lant or antiplatelet therapy before cataract surgery is associated with any increased risk of complications, as it would require a large study, on the order of 13,000 routine users of anticoagulants or antiplatelet therapy, to detect a 50% increase in medical events associated with discontinuation of therapy.

We examined the question of whether routine users of anticoagulant or antiplatelet therapy who continued their therapy before cataract surgery experienced different risks of ocular hemorrhage or hemorrhagic or thromboebolic events than did those who discontinued use of these therapies. We also compared the risk of these events with that in patients who were not routine users of these therapies in a large, prospective, randomized trial of the impact of preoperative medical testing for cataract surgery on adverse medical events in 19,283 cataract surgeries performed in the United States and Canada from June 1995 through June 1997.[18–21]

Materials and Methods

All patients undergoing cataract surgery at nine centers in the United States and Canada between June 1995 and June 1997 were eligible for enrollment in the study, as previously described.[18–21] Patients were excluded if they were under 50, they had a history of a myocardial infarction within the past 3 months, general anesthesia was planned, the surgery on a second eye was scheduled within 28 days of surgery on an enrolled first eye, informed consent could not be obtained, or the patient could not speak English or Spanish. Simple cataract surgeries and combined procedures were included.

Eligible patients were approached by study staff who explained the purpose and procedures of the study and sought written informed consent, including consent to obtain medical records related to adverse medical and ocular events that might be associated with surgery. Ethical approval for the study was obtained from the Joint Committee on Clinical Investigations of the Johns Hopkins University School of Medicine and from each clinical center's institutional review board.

Exposures

Information on pre-existing medical conditions and usual medication use was obtained from a prospective, systematically collected history and physical examination completed by a physician or physician's assistant at the time he/she cleared the patient for surgery. They also recorded whether routinely used medications including warfarin and aspirin were to be stopped before surgery. On the day of surgery, patients were asked whether warfarin or aspirin had been taken on the day of surgery and, if not, the number of days since the last dose. Patients' medication use was categorized into three groups: those who were not routine users and routine users who did or did not continue medication use within 4 days of surgery (warfarin) and 14 days of surgery (aspirin). This was done separately for any warfarin or any aspirin use. The number of patients who were not routine users but who reported use within the time frames given above was small, and they were included in the much larger group of nonusers.

Outcomes

Surgeries in this analysis included those that were completed and those that were canceled in the operating room. This allowed for the inclusion of cancellations in the operating room that were due to retrobulbar or peribulbar hemorrhage. If the surgery took place, the anesthesiologist or nurse anesthetist recorded any intraoperative medical events. If there was a delay of discharge, the reason for a delay was recorded by the nursing staff before discharge. Patients were telephoned 7 days after surgery. Deaths, hospitalizations, or unplanned visits to a health care provider were recorded. The medical records of all medical events identified at these times were reviewed by an internist and anesthesiologist to determine whether a medical event meeting the study definition had occurred. Medical events of possible relevance to warfarin or aspirin use included myocardial infarction, myocardial ischemia, stroke, transient ischemic attack (TIA), thromboembolic events, and ocular and other systemic hemorrhages. For the rate of retrobulbar hemorrhage alone, the denominator included only those surgeries performed under local anesthesia by injection.

Analysis

The analysis combined data from both treatment arms of the randomized trial because there was no impact of routine preoperative medical testing on adverse medical events (the primary trial outcome) during and after surgery.[19] There were also no differences in the rate of specific events such as stroke, TIAs, myocardial infarctions, or arrythmia. Patients assigned to no routine preoperative testing did get directed testing for any new or worsening problems identified by their internists. Rates of adverse medical and ocular events were calculated by dividing the number of events by the number of surgeries performed or canceled and expressed as events per 1000 surgeries. Rates were calculated for each of the three exposure groups separately. Relative risks (RRs) of adverse events and their associated 95% confidence intervals (CIs) were calculated using patients who were not routine users as the reference group. A second analysis compared routine users who did and did not continue medication use before surgery. Chi-square tests were used to calculate *P*s for comparisons of prevalence of pre-existing comorbid conditions between groups, and Fisher exact test (2 sided) was used to calculate *P*s for the comparison of incidence rates because the number of events was small. Analyses were done using the Statistical Analysis Software package (SAS Institute Inc., Cary, NC).

Results

A total of 19,354 patients undergoing 20,775 surgeries were approached to participate in the study. Agreement to participate was obtained for 19,584 surgeries (94.1%). Of these, 301 were canceled before patients reached the operating room and were never rescheduled during the study period. The remaining 19,283 surgeries were included in this analysis. Of these, 99.8% provided the final 7-day postsurgery interview. Retrobulbar anesthesia and/or peribulbar anesthesia were used in 14,183 surgeries (73.9%), with the remaining using topical local anesthesia without injection. Eighteen surgeries were canceled in the operating room; 12 were rescheduled during the study and 6 were not.

A history and physical examination were conducted before surgery, and data on medication use at the time of this examination were available for all except 303 surgeries (1.6%). Based on this examination, routine use of aspirin was most common in this population (22.4%). The indications for aspirin use and the dose were not recorded. There were 3.6% of patients using warfarin alone and 0.4% using both routinely (Table 1). Of the 4588 aspirin users, 13.8% were advised by their physician to stop usage before surgery. Among the 752 warfarin users, 10.5% were also advised to stop before surgery.

Table 1. Distribution of Routine Aspirin and Warfarin Use Obtained from a History and Physical Examination before Cataract Surgery

Medications	Routine Use	
	n	%
Aspirin	4517	23.8
Warfarin	681	3.6
Aspirin, warfarin	71	0.4
None	13,711	72.2
Total	18,980*	100.0

*303 missing medication use information on history and physical examination.

On the day of surgery, patients were asked whether they had taken aspirin in the past 14 days or warfarin in the past 4 days. Seventy-seven percent of surgeries involved patients who were not routine users of aspirin (Table 2). The percentage of surgeries in which the patient was a routine user of aspirin and had taken it within the past 14 days was 18.0%, and 5.2% were routine users who had discontinued use in the 2 weeks before surgery. Among routine users, 22.5% stopped aspirin in the 2 weeks before surgery, and 28.3% of warfarin users stopped in the 4 days before surgery. The proportion of patients reporting discontinuation of use before surgery was greater than that reported by physicians during the history and physical examination.

A number of pre-existing comorbid conditions, as reported by the physician on the history and physical examination, were associated with the use of anticoagulants and antiplatelet agents (Table 3). Aspirin use was significantly associated with a higher prevalence of angina, history of myocardial infarction, ventricular arrhythmia, coronary artery bypass surgery, TIA, and hypertension. The prevalence of bleeding disorders was statistically but not clinically significantly lower among those taking aspirin (0.2%) than among those not taking aspirin (0.4%). The prevalence of all the previously mentioned conditions, as well as other valvular disease, bleeding disorders, and diabetes was significantly higher among routine warfarin users than among other patients. As a general measure of operative risk, the American Society of Anesthesiologists classification of risk was used. Routine aspirin and warfarin users were more likely to be placed in class III or IV, indicating severe systemic disease that is not incapacitating or incapacitating systemic disease that is a constant threat to life. The vast majority of patients in this study were in class III.

Out of 19,283 surgeries, there were 3 hyphemas, 1 vitreous hemorrhage, and 6 retrobulbar hemorrhages. The retrobulbar hemorrhages led to the cancellation of surgery. Among the 14,322 surgeries in which patients were not routine aspirin users, there were 8 ocular hemorrhagic events (0.56/1000 surgeries; Table 4). The rate was similar for routine users of aspirin who continued its use within 2 weeks of surgery (0.59/1000). There were no such events among the 977 surgeries in which routine aspirin use was discontinued. The rates of retrobulbar hemorrhage among those who had injection local anesthesia were similar for nonusers (0.47/1000) and routine continuous users (0.42/1000). Routine continuous aspirin use did not alter the risk of any medical events except for a higher risk of myocardial ischemia and TIA relative to nonroutine users. There was no evidence of an increased risk of stroke, TIA, or thromboembolic events among routine users who discontinued use (1.02/1000 surgeries) compared with those who

Table 2. Distribution of Aspirin and Warfarin Patterns of Use after the Decision to Have Cataract Surgery

Medications	No Routine Use		Routine Use			
			Discontinued Use		*Continued Use*	
	N	%	*N*	%	*N*	%
Aspirin*	14,322	76.7	977	5.2	3363	18.0
Warfarin*	18,215	96.1	208	1.1	526	2.8

*621 (aspirin) and 334 (warfarin) surgeries are missing data on routine use or use before surgery.

Table 3. Prevalence of Pre-existing Comorbidity by Routine Use of Aspirin or Warfarin

Comorbidity	Aspirin		Warfarin	
	No [n (%)]	*Yes [n (%)]*	*No [n (%)]*	*Yes [n (%)]*
Angina	975 (6.8)	914 (19.9)*	1762 (9.7)	127 (16.9)*
MI	719 (5.0)	844 (18.4)*	1408 (7.7)	155 (20.6)*
MI in past 6 mos	14 (0.1)	30 (0.7)*	34 (0.2)	10 (1.3)*
Thromboembolic disease	287 (2.0)	116 (2.5)*	313 (1.7)	90 (12.0)*
Other valvular disease	397 (2.8)	148 (3.2)	445 (2.4)	100 (13.3)*
Ventricular arrhythmia	269 (1.9)	126 (2.7)*	352 (1.9)	43 (5.7)*
Coronary artery bypass surgery	405 (2.8)	666 (14.5)*	944 (5.2)	127 (16.9)*
Transient ischemic attack	574 (4.0)	569 (12.4)*	974 (5.3)	169 (22.5)*
Bleeding disorders	56 (0.4)	7 (0.2)*	53 (0.3)	10 (1.3)*
Hypertension	5941 (41.3)	2333 (50.9)*	7884 (43.3)	390 (51.9)*
Diabetes	1867 (13.0)	682 (14.9)*	2401 (13.2)	148 (19.7)*
ASA risk class III or IV	4561 (31.7)	2099 (45.8)*	6050 (33.2)	601 (81.1)*

ASA = American Society of Anesthesiologists, MI = myocardial infarction.
*$P < 0.05$.

Table 4. Incidence of Ocular and Medical Intraoperative and Postoperative Events by Aspirin Use before Cataract Surgery

	No Routine Use [n = 14,322 (10,555)]	Routine Use, Discontinued [n = 977 (774)]	Routine Use, Continued [n = 3363 (2407)]
Ocular hemorrhage			
Hyphema	3 (0.21)	0	0
Vitreous	0	0	1 (0.30)
Retrobulbar*	5 (0.47)	0	1 (0.42)
Total	8 (0.56)	0	2 (0.59)
Medical events			
Myocardial infarction	10 (0.70)	0	3 (0.89)
Hemorrhagic cystitis	1 (0.07)	0	0
Myocardial ischemia	12 (0.84)	1 (1.02)	14 (4.16)†
Stroke	9 (0.63)	1 (1.02)	1 (0.30)
TIA	0	0	4 (1.19)†
Deep vein thrombosis	2 (0.14)	0	0
Total stroke/TIA/thrombosis	11 (0.77)	1 (1.02)	5 (1.49)

TIA = transient ischemic attack.
*Incidence is cases per 1000 among injection anesthesia surgeries. Numbers in parentheses are numbers of injection anesthesia surgeries.
†$P < 0.05$ for comparison with no routine use.

Table 5. Incidence of Ocular and Medical Intraoperative and Postoperative Events by Warfarin Use before Cataract Surgery

	No Routine Use [n = 18,215 (13,434)]	Routine Use, Discontinued [n = 208 (174)]	Routine Use, Continued [n = 526 (348)]
Ocular hemorrhage			
Hyphema	3 (0.16)	0	0
Vitreous	1 (0.05)	0	0
Retrobulbar*	6 (0.45)	0	0
Total	10 (0.55)	0	0
Medical events			
Myocardial infarction	11 (0.60)	1 (4.81)	1 (1.90)
Hemorrhagic cystitis	1 (0.05)	0	0
Myocardial ischemia	26 (1.43)	0	3 (5.70)†
Stroke	10 (0.55)	0	1 (1.90)
TIA	4 (0.22)	0	0
Deep vein thrombosis	1 (0.05)	0	1 (1.90)
Total stroke/TIA/ thrombosis	15 (0.82)	0	2 (3.80)

TIA = transient ischemic attack.
*Incidence is cases per 1000 among injection anesthesia surgeries. Numbers in parentheses are numbers of injection anesthesia surgeries.
†$P < 0.05$ for comparison with no routine use.

continued its use (1.49/1000 surgeries) (RR = 0.7, 95% CI = 0.1–5.9). The RR of myocardial infarction or ischemia was 0.7 (95% CI = 0.1–5.9), comparing those who continued aspirin use with those who discontinued.

There were no ocular hemorrhages among any warfarin users, whether the use was routine or not and whether they continued or discontinued use within 4 days of surgery (Table 5). Routine warfarin users who continued use in the 4 days before surgery were at increased risk of myocardial ischemia (RR = 4.0, 95% CI = 1.2–13.2) compared with those not using warfarin routinely. There were no statistically significant differences in overall medical event rates between routine users who continued and discontinued warfarin use.

There were 7231 surgeries in the four clinical centers that recorded the results of a 24-hour ocular examination. Of these, 55% were done under injection local anesthesia. The rate of retrobulbar hemorrhages was 2.3 per 1000 surgeries. This is 3 times higher than the rate of 0.7/1000 obtained from all surgeries in the study using only the 7-day interview for ascertainment of ocular hemorrhagic events. Using the subset with 24-hour data, there were 2 retrobulbar hemorrhages out of 920 (2.2/1000) among routine aspirin users who continued its use and none in the 173 users who discontinued use. There was 1 retrobulbar hemorrhage out of 136 (7.4/1000) among warfarin users who continued its use and none among 36 users who discontinued its use. Among injection and topical local surgeries with 24-hour ocular examinations, the rate of hyphema, choroidal, vitreous, or lid hemorrhage was 1.4% among aspirin users who continued use and 1.6% among those who did not. The rate was 0.4% among warfarin users who continued use and 2.1% among those who discontinued use (1 event in each group).

Discussion

Rates of all adverse events were extremely low. This is safe surgery, despite the age of these patients (72 years on average) and the high rates of pre-existing comorbidities (10% with angina, 40% with hypertension, 8% with a history of myocardial infarction, and 14% with diabetes). Among routine users of anticoagulants or antiplatelet agents, there was no evidence to suggest that patients who continued use were at increased risk of ocular hemorrhagic events, nor that those who discontinued use were at increased risk of medical events for which these medications are routinely prescribed.

The strengths of this study are the large number of representative cataract surgeries and the systematic and prospective manner in which the exposures and events were collected and verified. The surgeries represented a mix of community hospitals and academic, medical, and private practice freestanding surgical centers within the United States and Canada. The age and comorbidity profile of these patients reflected the population of patients presenting for cataract surgery in the United States.[22] Medication use was collected from physicians at the time the patient was cleared for surgery, and use before surgery was obtained by the anesthesiologist or nurse anesthetist from the patient on the day of surgery. Intraoperative events were recorded by the anesthesiologist/nurse anesthetist immediately after surgery, and postoperative events were recorded before discharge or identified through a phone call to the patient within 7 days of surgery. Medical records surrounding these events were obtained, and all potential events were reviewed by a panel of internists and anesthesiologists who agreed on whether the study definitions of events were met.

The major drawbacks of this study were the rarity of events, which made it hard to detect differences between groups with varying usage of anticoagulants and antiplatelet agents, and the observational study design. Although event rates were low, there was some evidence for an increased risk of myocardial ischemia/angina and TIA among routine users of aspirin or warfarin who continued to use these before surgery, compared with those who were not routine users. This increased risk is likely explained by the higher prevalence of pre-existing comorbidities among routine aspirin and warfarin users. For example, a history of angina among routine aspirin and warfarin users was 2 to 3 times that of nonusers, and a history of TIA was 3 to 4 times more common among routine aspirin or warfarin users. There were clearly indications for use of aspirin or warfarin that put routine users at higher risk for these events than those not on these medications, but the event rates were so low that adjustment for pre-existing comorbidities was not feasible. A comparison of routine users who continued or discontinued medication likely reduced this selection bias, but there may have been some residual bias in that those who were told to continue use may have been considered at higher risk for the medical events for which the medication was prescribed than those who were allowed to discontinue medication before surgery. There is some evidence in these data that those who continued aspirin or warfarin use had slightly higher rates of certain medical events than did those who discontinued their use. There is no evidence that discontinuation of routine use resulted in higher medical event rates, but we cannot comment on what the rates of medical events for patients who stayed on anticoagulants might have been had we required them to stop usage before surgery, and there is no way

to adjust adequately for this possible selection bias. These data indicate how important a randomized, controlled trial design would be in addressing this question effectively.

The risk of ocular hemorrhages was very low, and there was no evidence to suggest that it was higher among those who continued rather than stopped routine anticoagulant use before surgery. Because the rate of thrombotic events was higher than that of ocular hemorrhagic events requiring cancellation of surgery in the study as a whole, this lends support to those who would continue antithrombotic therapy. However, the rate of retrobulbar hemorrhage ascertained from the subset of centers in which 24-hour ocular examination results were recorded was 3 times higher than that obtained from the overall study population, leading to a higher rate of ocular hemorrhages than that of thrombotic events. The rate of arterial thrombotic events was higher than the rate of venous thrombotic events. This lends support to the conventional wisdom among internists that it is safer to discontinue anticoagulant use in the setting where there is venous rather than arterial disease. If the internist feels that discontinuation is appropriate, these data suggest the timing of discontinuation to be 2 weeks for aspirin and 4 days for warfarin.

One concern not addressed in these analyses is whether there were patients scheduled for surgery who were told to discontinue anticoagulant or antiplatelet medication use and then had a thrombotic event that resulted in cancellation of surgery. Such events would be associated with discontinuation of anticoagulants/antiplatelet agents but not appear in these results because the surgery never took place. Records of cancellations for medical events were maintained in our study. There were 63 such events, 11 of them due to stroke, TIAs, myocardial infarction, or deep vein thrombosis. Through a baseline questionnaire, we did know whether these patients reported routine aspirin or warfarin use. Of the 11 events, 3 involved routine aspirin users, but none of them reported warfarin use. These events occurred before the completion of a history and physical examination, so it is unlikely that routine users would have stopped their medication. Hence this group is unlikely to be informative about the risks of discontinuation of anticoagulant medication.

The data from this study support the continued use of anticoagulants and antiplatelet agents among routine users because the risks of adverse medical events were not increased by discontinuation of these medications. Similarly, there was no evidence that continued use increased the risk of ocular hemorrhagic events, further supporting their continued use. There may be strong selection bias in the utilization pattern of these agents. Hence, the only way to answer this question definitively would be through a randomized controlled trial, which would require a sample size of more than 20,000 anticoagulant users.

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